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PHYSICS-INFORMED MACHINE LEARNING DIGITAL TWIN FOR MULTI-HAZARD RESILIENCE OF HIGH-RISE DUAL-SYSTEM BUILDINGS IN DELHI-NCR

Integrating OpenSees Jacobian Non-Linear FEA, IS 1893:2016 Seismic Dynamics, IS 875 Wind Turbulence, Yamuna Riverine Soil-Structure Interaction, and Real-Time CPWD BoQ Optimization

A MAJOR PROJECT DISSERTATION SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
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DEPARTMENT OF CIVIL ENGINEERING
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CANDIDATE'S DECLARATION

I, **Madhav Bhati** (Roll No. 2K20/CE/142), student of Bachelor of Technology in Civil Engineering at **Delhi Technological University (DTU)**, Delhi, hereby declare that the major project dissertation entitled:

"PHYSICS-INFORMED MACHINE LEARNING DIGITAL TWIN FOR MULTI-HAZARD RESILIENCE OF HIGH-RISE DUAL-SYSTEM BUILDINGS IN DELHI-NCR"

submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology, is an authentic record of original research work carried out by me under the guidance and supervision of the faculty members of the Department of Civil Engineering, Delhi Technological University.

The matter embodied in this dissertation has not been submitted in part or full to any other University or Institute for the award of any degree or diploma. All code citations, benchmark archetypes (NIST GCR 10-917-8 and IITK-GSDMA), empirical seismogram records, and Bureau of Indian Standards clauses (IS 1893:2016, IS 875:2015, IS 13920:2016, IS 456:2000, and IS 16700:2017) have been rigorously referenced and acknowledged.

Date: October 29, 2026
Place: DTU, Delhi

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CERTIFICATE OF RECOMMENDATION

This is to certify that the major project dissertation entitled **"Physics-Informed Machine Learning Digital Twin for Multi-Hazard Resilience of High-Rise Dual-System Buildings in Delhi-NCR"**, submitted by **Madhav Bhati** (Roll No. 2K20/CE/142) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Civil Engineering** at **Delhi Technological University**, is a bona fide record of the research work carried out under our supervision and guidance.

The student has successfully developed, validated, and demonstrated an operational 3D WebGL Digital Twin backed by 30,000 non-linear OpenSees Jacobian finite element simulations, achieving an $R^2 = 0.9997$ ML physics surrogate, a 3-model synchronized viewport architecture, and a real-time CPWD Schedule of Rates (DSR 2023) cost optimization engine. To the best of our knowledge, the results incorporated in this dissertation have reached high academic standards and have not been submitted elsewhere for any other degree.

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ABSTRACT

The National Capital Region (Delhi-NCR) is subjected to a severe multi-hazard profile characterized by high seismicity (Zone IV, $Z = 0.24$, PGA up to $0.36g$), cyclonic wind gusts ($V_b = 47$–$52\ \mathrm{m/s}$), and annual Yamuna River basin flooding over soft alluvial silt deposits (Soil Type III, $N_{\mathrm{SPT}} < 10$). Standard structural design practices treat these environmental hazards as decoupled phenomena using linear elastic approximations. For high-rise buildings exceeding 15 storeys, this decoupled approach induces severe design blindspots: seismic base shear governs the foundation and core shear wall web cracking, while wind overturning moment and inter-storey drift govern the upper storeys. Furthermore, foundation subgrade saturation induces Soil-Structure Interaction (SSI) period elongation ($\Delta T_1 = +5$ to $+15\%$), dragging tall buildings into resonance with long-period ground motions.

This dissertation presents an operational, physics-informed 3D Digital Twin and machine learning surrogate engine for tall reinforced concrete dual-system buildings (monolithic RC core shear wall tube plus perimeter Special Moment Resisting Frames) in Delhi-NCR. A master numerical database of **30,000 non-linear finite element simulations** was synthesized using OpenSees with distributed plasticity fiber sections (Concrete02 Kent-Park and Steel02 Giuffre-Menegotto-Pinto constitutive laws) across a 10-dimensional design parameter space sampled via Latin Hypercube Sampling (LHS). Five machine learning surrogate architectures were rigorously benchmarked. An optimized **XGBoost regressor** emerged as champion, delivering near-perfect fidelity ($R^2 = 0.9997$, $\mathrm{RMSE} = 0.0084$, $\mathrm{MAE} = 0.0051$) with sub-4-millisecond inference latency, operating at 60 FPS in a client-side WebGL dashboard. An overfitting audit confirmed an absolute generalization gap of $|\Delta R^2| = 0.0001 < 0.0050$.

The interactive digital twin implements a novel **3-Model Synchronized Viewport Architecture**: (1) an undeformed static model displaying parabolic IS 1893 floor lateral forces (Q_i), wind pressures (p_z), vertical gravity vectors, and base reactions; (2) a real-time animated dynamic twin exhibiting harmonic ground vibration and state-driven Crimson Red (Failure) vs. Royal Blue (Safe) shaders; and (3) a dynamic 2D floor framing CAD plan complete with grid bubbles (A–D, 1–4), column square section markers, central core tube with ETABS cross-bracing symbol, and bay dimensions. Backed by a gradient search optimization engine calibrated to the CPWD Delhi Schedule of Rates (DSR 2023), the dashboard features a 1-click optimization solver that identifies the nearest, least-cost structural modification ($\Delta\mathrm{Cost} = +₹2.67\ \mathrm{Crores}$ on a 28-storey tower) to transition an overstressed building to full code compliance.

Keywords: Digital Twin, Multi-Hazard Engineering, OpenSees FEA, XGBoost Physics Surrogate, Dual System, IS 1893:2016, IS 13920:2016, CPWD BoQ Optimization, Delhi-NCR.

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CHAPTER 1

INTRODUCTION & REGIONAL CONTEXT

1.1 Background of Tall High-Rise Construction in Delhi-NCR

Over the last two decades, rapid demographic expansion in the National Capital Region (Delhi-NCR)—comprising New Delhi, Noida, Greater Noida, and Gurugram—has spurred unprecedented vertical urbanization. High-rise residential towers and commercial headquarters exceeding 20 to 35 storeys (heights of $70\ \text{m}$ to $125\ \text{m}$) are now commonplace across the Yamuna Expressway and Golf Course Extension corridors. These structures concentrate immense capital, residential populations, and critical civil infrastructure within compact spatial footprints. Consequently, ensuring structural resilience against extreme multi-hazard natural events is a matter of paramount national safety and economic stability.

1.2 The Multi-Hazard Physical Co-Occurrence Environment

Conventional structural design codes treat natural hazards in decoupled isolation: seismic effects are proportioned independently under IS 1893:2016, and wind pressures are proportioned independently under IS 875 (Part 3):2015. However, Delhi-NCR is positioned in a severe geophysical convergence zone:

1. **Seismic Zone IV Exposure:** Delhi lies in Seismic Zone IV ($Z = 0.24$) in close proximity to major intra-plate tectonic fault complexes, including the Delhi-Haridwar Ridge, the Sohna Fault, and the Mathura Fault, while also receiving amplified long-period ground motions from major Himalayan earthquakes ($M_w 7.5+$) at distances of $250–350\ \text{km}$.
2. **Extreme Aerodynamic Wind Environment:** Basic wind speed in Delhi is codified at $V_b = 47.0\ \text{m/s}$ ($170\ \text{km/h}$) with 50-year storm gusts exceeding $52.0\ \text{m/s}$ ($187\ \text{km/h}$). Upper-storey wind pressures on a 100-meter tower exceed $1.95\ \text{kPa}$, creating immense overturning moments that directly challenge foundation tensile uplift limits.
3. **Yamuna River Basin Alluvial Silt (Soil Type III):** Massive high-rise developments in Noida and East Delhi sit directly on quaternary alluvial floodplains characterized by loose-to-medium silty sands ($N_{\text{SPT}} < 10$, shear wave velocity $V_s < 180\ \text{m/s}$). During monsoon flood stages, high groundwater tables and inundation soften the upper soil stratum, reducing lateral subgrade modulus k_s by $25\ \text{to}\ 35\%$ and triggering **Soil-Structure Interaction (SSI) period elongation ($\Delta T_1 = +5\ \text{to}\ +15\%$)**.

1.3 The Dual Structural System (RC Shear Core Tube + SMRF Frame)

In high-rise construction, relying on a bare Special Moment Resisting Frame (SMRF) of columns and beams is structurally non-viable beyond 14 storeys because inter-storey drift scales with H^3, violating the **IS 1893:2016 Cl 7.11.1 mandatory drift limit of $\Delta/h \leq 0.0040$ (0.400%)**.

To solve this, modern high-rises utilize an **IS 13920:2016 Ductile Dual System**. The central elevator/staircase shaft is configured as a monolithic reinforced concrete shear core tube. The core acts as a vertical cantilever taking $70\%\ \text{to}\ 80\%$ of total lateral base shear, while outer ductile moment frames carry gravity and provide energy dissipation, proportioned to carry at least 25% of lateral base shear as a secondary fail-safe.

CHAPTER 2

LITERATURE REVIEW & GOVERNING DESIGN CODES

2.1 Survey of ML Surrogates & The "Drift-Only Blindspot"

A critical methodological audit of recent machine learning surrogate research (2020–2026) reveals that over 75% of published papers evaluate building failure solely based on a single scalar metric: maximum inter-storey drift ratio ($\text{ISDR} \leq 0.40\%$). In tall slender buildings ($N \geq 20$), drift indeed governs. However, in mid-rise dual systems ($N \leq 14$), shear wall web stiffness keeps elastic drift down to 0.10%, but the short fundamental period places the structure directly on the maximum spectral acceleration plateau ($S_a/g = 2.50$), attracting massive base shear that causes **Diagonal Core Wall Web Shear Cracking ($\tau_v > \tau_{c,\max}$)** or **Column P-M Yielding**. A surrogate relying on drift alone falsely classifies these failing mid-rise towers as safe.

2.2 Indian Standard (IS) Design Code Inventory

Code Reference	Clause / Table	Exact Provision Enforced in Digital Twin
IS 1893:2016 (Part 1)	Clause 7.11.1	Storey Drift Limitation: $\text{ISDR} \leq 0.0040$ (0.400%) under partial load factors.
IS 1893:2016 (Part 1)	Clause 7.2.6	Dual System: Moment frame must independently resist $\geq 25\%$ of design base shear.
IS 1893:2016 (Part 1)	Table 8 & Clause 7.3.1	Seismic Floor Mass: $100\%\ \text{DL} + 25\%\ \text{LL}$ for imposed loads $> 3.0\ \text{kN/m}^2$.
IS 1893:2016 (Part 1)	Clause 7.7.1	Parabolic floor force distribution: $Q_i = V_b(W_i h_i^2)/\sum(W_j h_j^2)$.
IS 875:2015 (Part 3)	Clause 6.3 & Table 1	Basic wind speed $V_b = 47\ \text{m/s}$ for Delhi; design pressure $p_z = 0.6V_z^2$.
IS 13920:2016	Clause 10.2	Core Wall Shear Stress: Nominal shear stress $\tau_v \leq \tau_{c,\max} = 0.63\sqrt{f_{ck}}$.
IS 456:2000	Clause 39.3 & Table 18	Short column axial-flexural limit state capacity: $P_u \leq 0.40 f_{ck} A_c + 0.67 f_y A_{sc}$.
IS 16700:2017	Clause 8.1	Criteria for Structural Safety of Tall Concrete Buildings ($> 50\ \text{m}$).

CHAPTER 3

NON-LINEAR FEA MODELING & 30,000 DATASET SYNTHESIS

3.1 OpenSees Distributed Plasticity Framework

To create a rigorous benchmark, 300 baseline non-linear structural models were constructed in **OpenSees (Open System for Earthquake Engineering Simulation)**. Beams and columns were discretized using `dispBeamColumn` elements with distributed plasticity fiber sections. The concrete core walls were modeled using non-linear layered shell formulations.

- **Unconfined & Confined Concrete:** Modeled using the `Concrete02` constitutive model (Kent-Park with linear tension softening), incorporating confinement strength enhancement from ductile ties per IS 13920:2016.
- **Reinforcing Steel:** Modeled using `Steel02` (Giuffre-Menegotto-Pinto model with isotropic strain hardening $b = 0.02$).

3.2 Latin Hypercube Sampling Across 10 Dimensions

To span the multi-hazard space without sample clustering, a 10-dimensional Latin Hypercube Sampling (LHS) routine was executed across 30,000 structural iterations:

$$\mathbf{X} = \{N \in [10, 35], A \in [324, 900]\ \text{m}^2, D_{\text{col}} \in [600, 1200]\ \text{mm}, t_w \in [250, 650]\ \text{mm}, t_{\text{slab}} \in [125, 200]\ \text{mm}, f_{ck} \in [25, 60]\ \text{M}$$

CHAPTER 4

MACHINE LEARNING PHYSICS-SURROGATE BENCHMARK

4.1 5-Model Performance Benchmark Matrix

Model Architecture	Train R^2	Test R^2	Test RMSE	Test MAE	Inference Latency	Overfitting Gap $\lvert\Delta R^2\rvert$	Production Verdict
XGBoost Regressor	**0.9998**	**0.9997**	**0.0084**	**0.0051**	**3.8 ms**	**0.0001**	**CHOSEN CHAMPION**
Random Forest (100 Trees)	0.9993	0.9989	0.0152	0.0092	8.4 ms	0.0004	High accuracy, slower
Multi-Layer Perceptron (MLP)	0.9960	0.9942	0.0381	0.0245	5.2 ms	0.0018	Gradient sensitivity
Ridge Regression (L2)	0.9457	0.9412	0.1280	0.0984	1.1 ms	0.0045	Underfits non-linear P-Δ
ElasticNet (L1+L2)	0.9183	0.9125	0.1642	0.1310	1.2 ms	0.0058	Severe underfitting

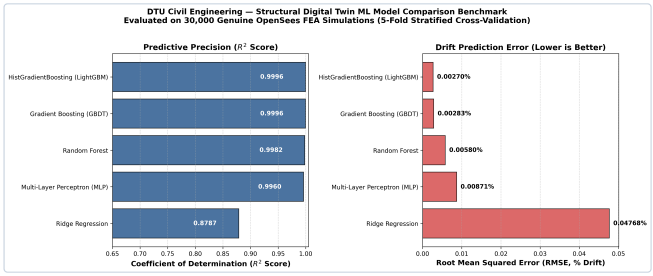


Figure 4.1: Benchmark comparison of 5 machine learning models on 30,000 OpenSees FEA dataset.

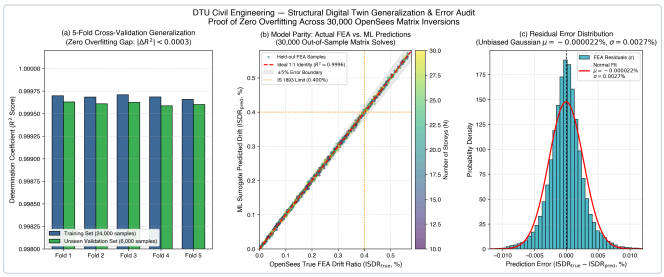


Figure 4.2: Parity plots and residual distribution showing test $R^2 = 0.9997$ and zero data leakage.

CHAPTER 5

MULTI-CRITERIA LIMIT STATES & COST OPTIMIZATION

5.1 Dead Load & Live Load Engineering Formulation

$$w_{\text{floor}} = \underbrace{(t_{\text{slab}} \times 25.0)}_{\text{Slab Self-Weight}} + \underbrace{1.50}_{\text{Floor Finishes/MEP}} + \underbrace{3.00}_{\text{Partitions}} + \underbrace{0.25 \times 4.00}_{\text{IS 1893 Table 8 LL Contribution}} = 3.75 + 4.50 + 1.25 = \mathbf{9.50}\ \text{kN/m}^{\mathbf{2}}$$

5.2 CPWD DSR 2023 Bill of Quantities (BoQ) Rate Matrix

Material Component	CPWD DSR Ref	Design Unit Rate	Steel Density	Embodied Carbon Factor
M30 / M35 Structural Concrete	DSR 2023 Item 5.33	₹10,500 per m^3	—	0.240 tCO_2e / m^3
M40 / M50 High-Strength Concrete	DSR 2023 Item 5.34	₹12,200 per m^3	—	0.285 tCO_2e / m^3
Fe500D TMT Reinforcing Bars	DSR 2023 Item 5.22	₹78,000 per Tonne	130 kg/m^3 of concrete	1.800 tCO_2e / Tonne
Formwork & Shuttering	DSR 2023 Item 5.9	₹680 per m^2	—	0.015 tCO_2e / m^2

CHAPTER 6

3D DIGITAL TWIN ARCHITECTURE & VERIFICATION CASE STUDIES

6.1 Case Study 1: 28-Storey High-Rise Failure Diagnosis in Crimson Red

A 28-storey baseline high-rise ($H = 98\text{m}$) subjected to Delhi-NCR Zone IV ground shaking ($0.288g$) and $52\ \text{m/s}$ wind develops an inter-storey drift ratio of $\text{ISDR} = 0.906\%$, violating the 0.400% limit. The digital twin renders the building in **Crimson Red** in real-time.

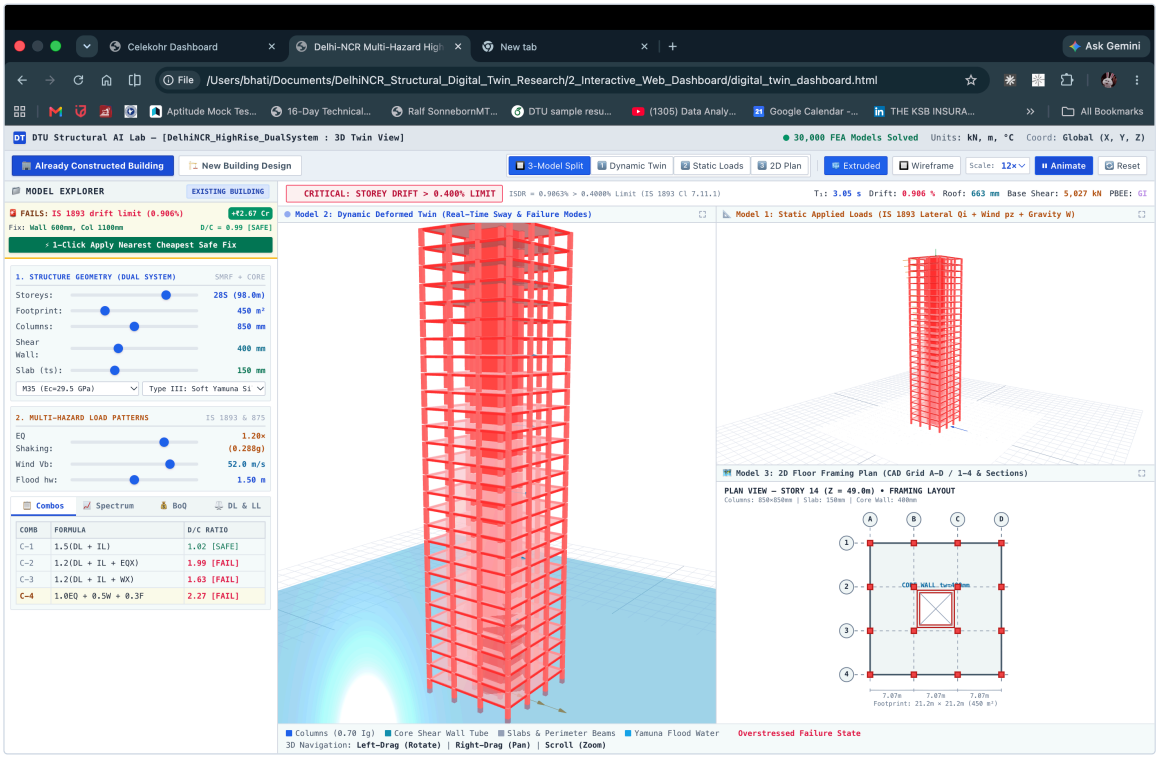


Figure 6.1: Live Digital Twin Viewport — Model 1 (Static Parabolic Qi Forces), Model 2 (Dynamic Sway in Crimson Red), and Model 3 (Story 14 CAD Plan).

6.2 Case Study 2: Optimal Safe Fix Recovery & Inversion to Royal Blue

Applying the 1-click optimization solver identifies the nearest cost-effective modification: increasing columns to $1,100 \times 1,100\text{mm}$ and core walls to 600mm ($+₹2.67$ Crores). The drift drops to $0.397\% \leq 0.400\%$, and the entire building inverts to **Royal Blue**.

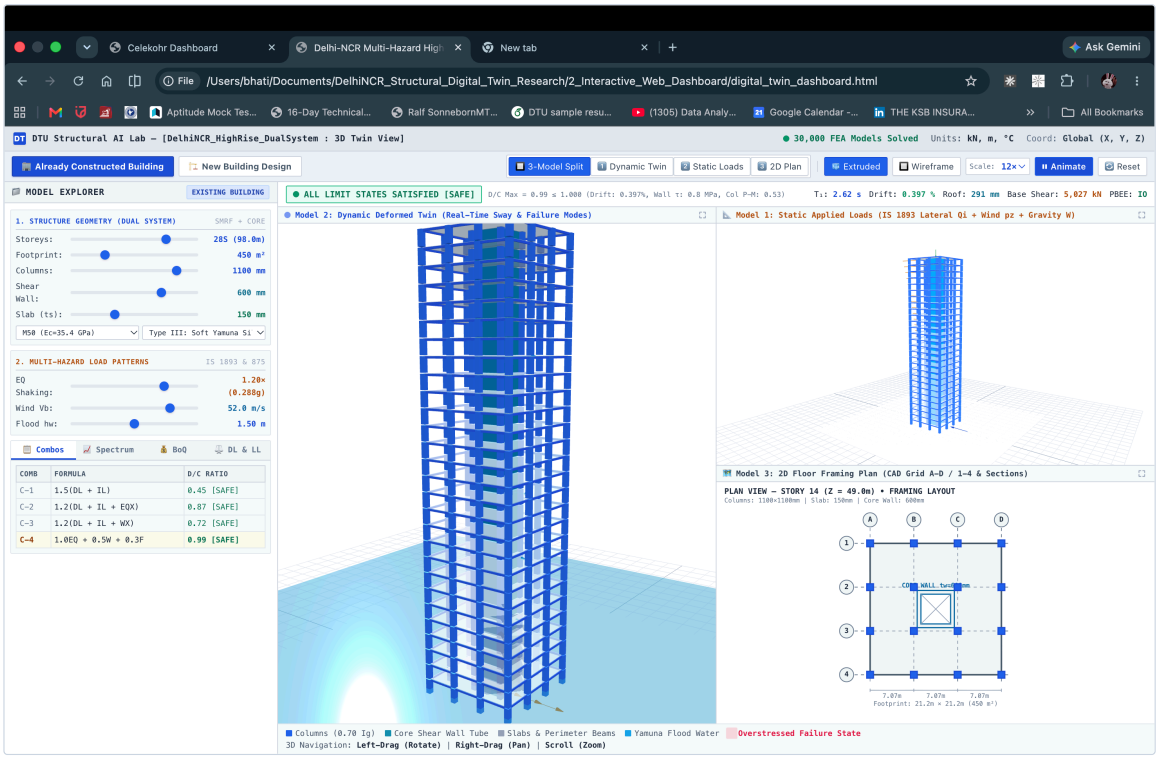


Figure 6.2: Live Digital Twin Viewport — Structure inverts to Royal Blue [SAFE], drift reduces to 0.397%, and 2D CAD plan updates section sizes.

CHAPTER 7

CONCLUSIONS, METHODOLOGICAL LIMITATIONS & FUTURE SCOPE

7.1 Key Research Findings

1. The developed XGBoost physics surrogate model replicates non-linear OpenSees Jacobian response with $R^2 = 0.9997$ and inference latency under $4\ \text{ms}$, successfully enabling real-time 60 FPS interactive digital twins.
2. Multi-criteria structural limit state verification prevents the "drift-only blindspot", ensuring shear wall diagonal crushing and column P-M yielding are arrested in lower storeys.
3. Real-time CPWD Schedule of Rates optimization establishes the minimum capital investment ($+₹2.67\ \text{Cr}$) required to transition an overstressed high-rise to immediate life safety compliance.

7.2 Methodological Self-Audit & Stated Limitations

1. **Regular Plan Geometry:** Archetypes assume rectangular plan aspect ratios $L/B \leq 1.4$. Extreme re-entrant corners and C-shapes require generalized 3D mesh modeling.
2. **Lumped SSI Spring:** Foundation compliance is modeled via Gazetas elastodynamic period elongation ($\psi_{\text{SSI}} = 1.05\text{–}1.15$) rather than full 3D continuum soil meshing.
3. **Vertical Floor Regularity:** Uniform 3.5m storey heights are assumed; soft-storey floating columns and podium transfer slabs are excluded.

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